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Helmet

Abstract

The helmet according to an embodiment of the present invention includes a helmet body; a pad coupled to the inside of the helmet body, and a coupling member that separably couples the helmet body and the pad, wherein the coupling member may include a male coupling member disposed in one of either the inside of the helmet body or the pad, and a female coupling member disposed in the other of either the inside of the helmet body or the pad and into which the male coupling member is inserted and fixed, wherein the male coupling member has a first inclined surface at a portion inserted and fixed into the female coupling member, and the female coupling member has a second inclined surface corresponding to the first inclined surface, and when an external force is applied to the pad in a first direction so that the male coupling member is separated from the female coupling member in a state in which the male coupling member and the female coupling member are coupled, the first inclined surface moves along the second inclined surface, and the pad is separated from the helmet body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase Application under 35 U.S.C. 371 of International Application No. PCT/KR2022/004942, filed on Apr. 6, 2022, which claims priority to and the benefit of Korean Patent Application No. 10-2021-0045327, filed on Apr. 7, 2021, the disclosures of which are incorporated herein by reference in their entirety.

FIELD

(2) The present invention relates to a helmet, and more particularly, to a helmet capable of removing a pad coupled to the inside of the helmet in emergency situations.

BACKGROUND

(3) In the case of a two-wheeled vehicle, there is a very high risk that the shock will be transmitted to a rider in the event of an accident. Accordingly, it is essential for the rider of the two-wheeled vehicle to wear a riding safety helmet, that is, a helmet, to protect his head.

- (4) In general, the helmet includes an outer shell that primarily protects the wearer's head from external impact, and a pad that is coupled to the inside of the outer shell. In this way, the pad provided inside the outer shell is in close contact with the head of the wearer to provide a sense of wearing stability and prevent the helmet from being easily peeled off when wearing the helmet.
- (5) Meanwhile, when an accident occurs and rescue is required for the wearer, the helmet must be removed from the wearer's head. However, it is difficult to separate the helmet due to the pad closely attached to the wearer's head and secondary injuries occur in the wearer's neck during the process of forcibly separating the helmet.
- (6) Therefore, there is a need to develop a helmet that can easily remove the pad in an emergency so that the pad can be stably fixed inside the helmet to provide a comfortable fit while easily separating the helmet from the wearer's head in the event of an accident.
- (7) (Priority Document) Patent No. 1995947, "Helmet being capable of removing pad in case of emergency" issued by the Korean Intellectual Property Office on Jul. 3, 2019.

SUMMARY

- (8) The present invention is to solve the problems of the prior art. The purpose of the present invention is to provide a helmet that is easily separated from the wearer's head in an emergency because the pad that is coupled to the inside and disposed between the outer shell and the wearer's head can be effectively removed by a force having a specific direction.
- (9) According to an aspect of the present invention, there is provided a helmet including a helmet body; a pad coupled to the inside of the helmet body; and a coupling member that separably couples the helmet body and the pad, wherein the coupling member may include a male coupling member disposed in one of either the inside of the helmet body or the pad, and a female coupling member disposed in the other of either the inside of the helmet body or the pad and into which the male coupling member is inserted and fixed, wherein the male coupling member has a first inclined surface at a portion inserted and fixed into the female coupling member, and the female coupling member has a second inclined surface corresponding to the first inclined surface, and when an external force is applied to the pad in a first direction so that the male coupling member is separated from the female coupling member in a state in which the male coupling member and the female coupling member are coupled, the first inclined surface moves along the second inclined surface, and the pad is separated from the helmet body.
- (10) In this case, the male coupling member may include a base portion fixed to either the inside of the helmet body or the pad; an extension portion that protrudes and extends from one side of the base portion and has the first inclined surface on one side thereof; and an expansion end portion provided at an end of the extension portion in an extension direction, at least a portion thereof extending outward than the extension portion in a direction perpendicular to the extension direction.
- (11) In addition, a portion of the extension portion having the first inclined surface may protrude outward than the expansion end portion in a direction perpendicular to the extension direction.
- (12) In addition, the helmet may further include a cover plate disposed to cover one surface of either the inside of the helmet body or the pad, and fixing the base portion.
- (13) In addition, the cover plate may have a through hole formed in a size that does not pass through the base portion, and wherein the base portion is disposed inside the helmet body or between the pad and the cover plate, and the extension portion extends through the through hole.
- (14) In addition, the extension portion may include a first extension portion extending through the through hole, and a second extension portion disposed between the first extension portion and the expansion end portion and having the first inclined surface on one side thereof.
- (15) In addition, the helmet may further include a movement prevention member may include a fixing portion fixed to the cover plate, and a movement prevention portion connected to the fixing portion to prevent the movement of the male coupling member by being fastened to the extension portion.

- (16) In addition, the second extension portion may include a fastening portion to which the movement prevention member is coupled, and the movement prevention portion is coupled to the fastening portion.
- (17) In addition, the female coupling member may further include a body having a receiving space to receive at least a portion of the extension portion and the expansion end portion and having the second inclined surface on one side of the inner wall of the receiving space, and an opening formed in the body in communication with the receiving space.
- (18) In addition, the opening may include a first portion provided in a shape surrounding the extension and a second portion connected to the first part and provided in a shape of a slot exposing the second inclined surface.
- (19) In addition, the second inclined surface and the second portion may be formed to be wider than the expansion end portion.
- (20) In addition, the helmet further includes a pulling portion provided to the pad to be pulled by the external force in the first direction, and wherein the first inclined surface and the second inclined surface may be oriented toward the pulling portion or a virtual point extending in the first direction.
- (21) According to the present invention, when an external force is applied in a specific direction according to the interaction between the first inclined surface and the second inclined surface, the pad is easily separated from the inside of the helmet body, thereby making it possible to safely separate the helmet from the wearer's head in an emergency.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a helmet according to an embodiment of the present invention.
- (2) FIG. 2 is an exploded perspective view of a helmet according to an embodiment of the present invention.
- (3) FIG. 3 is a view showing the fastening state between a helmet body and a pad in a helmet according to an embodiment of the present invention,
- (4) FIG. 4 is an exploded view of FIG. 3.
- (5) FIG. 5 is a perspective view of a male coupling member of a coupling member of a helmet according to an embodiment of the present invention.
- (6) FIG. 6 is a view showing a fastening state between a male coupling member and a female coupling member of a coupling member of a helmet according to an embodiment of the present invention.
- (7) FIG. 7 is a diagram showing a process in which the first inclined surface of the male coupling member of the coupling member of the helmet moves along the second inclined surface of the female coupling member to separate the male coupling member from the female coupling member according to an embodiment of the present invention.
- (8) FIG. 8 is a diagram showing a state in which the male coupling member of the coupling member of the helmet is separated from the female coupling member according to an embodiment of the present invention.
- (9) FIG. 9 is a cross-sectional view showing a part of a fastening state between a male coupling member and a female coupling member of a coupling member of a helmet according to an embodiment of the present invention.
- (10) FIG. 10 is a diagram showing the arrangement and orientation of the female coupling member of the coupling member of the helmet according to an embodiment of the present invention.
- (11) FIG. 11 is a diagram showing a method of removing a pad in a helmet during an emergency according to an embodiment of the present invention.

DETAILED DESCRIPTION

(12) Hereinafter, embodiments of the present disclosure will be described in detail so that those skilled in the art to which the present disclosure pertains can easily carry out the embodiments. The present disclosure may be implemented in many different forms and is not limited to the embodiments described herein. In order to clearly describe the present disclosure, portions not related to the description are omitted from the accompanying drawings, and the same or similar components are denoted by the same reference numerals throughout the specification.

(13) In the specification, it should be understood that the terms such as “comprise” or “have” are intended to specify the presence of features, numbers, steps, operations, components, parts, or combinations thereof described in the specification and do not preclude the possibility of the presence or addition of one or more other features, numbers, steps, operations, components, parts, or combinations thereof.

(14) In this specification, “front”, “rear”, “upper”, or “lower” and the like, which are spatially relative terms, may be used to describe the correlation with the components illustrated in the drawings. These are relative terms determined based on the figures and the positional relationship may be interpreted as opposite depending on the orientation. In addition, the fact that a component is “connected” to another component includes not only directly connected to each other but also indirectly connected to each other unless otherwise specified.

(15) FIG. 1 is a perspective view of a helmet according to an embodiment of the present invention. In addition, FIG. 2 is an exploded perspective view of a helmet according to an embodiment of the present invention.

(16) Referring to FIGS. 1 and 2, the helmet according to an embodiment of the present invention includes a helmet body **100**, a pad **200** coupled to the inside of the helmet body **100**, and a coupling member **300** that separably couples the helmet body **100** and the pad **200**.

(17) The helmet body **100** serves to protect the head of the wearer. The helmet body **100** may be made of a material capable of absorbing impact. For example, the helmet body **100** may include an outer shell **110** that is made of a hard synthetic resin or the like and has high strength, and a shock absorber **120** provided inside the outer shell **110**. In this case, the shock absorber **120** may be made of expanded polystyrene (EPS) or the like.

(18) In an embodiment of the present invention, the pad **200** may be coupled to the inner wall of the shock absorber **120** by the coupling member **300**. However, the helmet body **100** does not necessarily include the shock absorber **120**, the pad **200** does not also have to be coupled to the inner wall of the shock absorber **120**, and may be coupled to any other portion of the interior of the helmet body **100**.

(19) The helmet body **100** may have an opening **130** opened in front thereof to secure the user's view. In addition, the opening **130** may be opened and closed by the shield **140** rotatably coupled to the helmet body **100**.

(20) The pad **200** is disposed between the inside of the helmet body **100** and the wearer's head to provide a stable fit to the wearer. In an embodiment of the present invention, the pad **200** may be detachable coupled to the inside of the helmet body **100** (for example, the inner wall of the shock absorber **120**) by the coupling member **300**. The pad **200** coupled to the inside of the helmet body **100** may be in contact with the jaw, cheeks, ears, etc. of the helmet wearer.

(21) The pad **200** may be formed to have elasticity to perform a buffering action between the wearer's head and the helmet body **100** and to be deformed according to the shape of the wearer's head. In addition, the cover plate **400** may be disposed to cover one surface of either the inside of the helmet body **100** or the pad **200**. In an embodiment of the present invention, a cover plate **400** is disposed on one surface (a surface in contact with the helmet body **100**) of the pad **200**. That is, the cover plate **400** may cover one surface of the pad **200** and be disposed between the inner surface of the helmet body **100** and one surface of the pad **200**.

(22) The coupling member **300** includes a male coupling member **310** disposed in one of either the

inside of the helmet body **100** or the pad **200**, and a female coupling member **320** disposed in the other of either the inside of the helmet body **100** or the pad **200** and into which the male coupling member **310** is inserted and fixed.

(23) FIG. **3** is a view showing the fastening state between a helmet body and a pad in a helmet according to an embodiment of the present invention, and FIG. **4** is an exploded view of FIG. **3**. In addition, FIG. **5** is a perspective view of a male coupling member of a coupling member of a helmet according to an embodiment of the present invention, and FIG. **6** is a view showing a fastening state between a male coupling member and a female coupling member of a coupling member of a helmet according to an embodiment of the present invention. In FIG. **6**, illustrations of the helmet body **100**, the pad **200**, and the cover plate **400** are omitted.

(24) Referring to FIGS. **3** to **6**. The male coupling member **310** has a first inclined surface **311** at a portion inserted and fixed into the female coupling member **320**, and the female coupling member **320** has a second inclined surface **321** corresponding to the first inclined surface **311**. When an external force is applied to the pad **200** in a first direction so that the male coupling member **310** is separated from the female coupling member **320** in a state in which the male coupling member **310** and the female coupling member **320** are coupled, the first inclined surface **311** moves along the second inclined surface **321** to release the coupling of the male coupling member **310** and the female coupling member **320**, thereby easily separating the pad **200** from the helmet body **100**.

(25) The male coupling member **310** may include a base portion **312** fixed to either the inside of the helmet body **100** or the pad **200**, an extension portion **313** that protrudes and extends from one side of the base portion **312** and has a first inclined surface **311** on one side thereof, and an expansion end portion **314** provided at an end of the extension portion **313** in an extension direction, at least a portion thereof extending outward than the extension portion **313** in a direction perpendicular to the extension direction.

(26) In an embodiment of the present invention, the base portion **312** is fixed to the pad **200** side by the cover plate **400**. For example, the base portion **312** may be made of a plate, the cover plate **400** may include a through hole **410** formed in a size that does not pass through the base portion **312** and the base portion **312** may be disposed between the pad **200** and the cover plate **400**.

Meanwhile, according to another embodiment in which the male coupling member **310** is provided on the side of the helmet body **100**, the base portion **312** may be fixed inside the helmet body **100**.

(27) The extension portion **313** may extend through the through hole **410** (see FIGS. **3** and **4**).

More specifically, the extension portion **313** may include a first extension portion **313a** extending through the through hole **410** and a second extension portion **313b** disposed between the first extension portion **313a** and the expansion end portion **314** and having a first inclined surface **311** on one side thereof.

(28) The first extension portion **313a** may have a shape corresponding to the shape of the through hole **410** and may be extended from one surface of the base portion **312**. If the first extension portion **313a** and the through hole **410** have a corresponding shape, it may be helpful for the male coupling member **310** to be stably fixed to the cover plate **400**.

(29) The second extension portion **313b** may include a fastening portion **3131** to which the movement prevention member **500** is coupled and an additional extension portion **3132** which is additionally extended from the fastening portion **3131** in the extension direction and has the first inclined surface **311** on one side thereof. In this case, the first inclined surface **311** may be formed to be inclined inward from the outside in a direction perpendicular to the extending direction along the extending direction of the extension portion **313**. In addition, the coupling portion **3131** of the second extension portion **313b** is formed to have a cross-sectional area smaller than that of the first extension portion **313a** and the additional extension portion **3132** of the second extension portion **313b**, so the movement of the movement prevention member **500** may be restricted in the extension direction by the first extension portion **313a** and the additional extension portion **3132** in a state in which the movement prevention member **500** is coupled to the fastening portion **3131**.

(30) In an embodiment of the present invention, the male coupling member **310** is fixed to the pad **200** side, and since the pad **200** has elasticity, the male coupling member **310** may be pressed toward the pad **200** side and may move when the male coupling member **310** and the female coupling member **320** are intended to be coupled. Accordingly, smooth coupling between the male coupling member **310** and the female coupling member **320** may be hindered. The movement prevention member **500** is disposed to prevent the movement of the male coupling member **310**.

(31) The movement prevention member **500** may include a fixing portion **510** fixed to the cover plate **400** and a movement prevention portion **520** connected to the fixing portion **510** to prevent the movement of the male coupling member **310** by being fastened to the extension portion **313**. More specifically, the fixing part **510** may be formed of a hook fastened to the hook hole **420** formed in the cover plate **400**, and the movement prevention portion **520** may be formed of a C-ring which is fastened while surrounding the fastening portion **3131**.

(32) Of course, the configuration of the movement prevention member **500** is not limited by this, and the fastening structure between the fixing portion **510**, the movement prevention portion **520**, and the fastening portion **3131** fixed to the cover plate **400** can be modified in various forms. For example, the movement prevention portion **520** may be made of a male fastening portion, and the fastening portion **3131** may be made of a female fastening portion.

(33) The expansion end portion **314** is fixed in a receiving space **S** of the female coupling member **320** while the male coupling member **310** is fastened to the female coupling member **320**. The expansion end portion **314** may have a shape that interferes with the opening **323** of the female coupling member **320** when force is not applied in the first direction and is applied in the other direction so that separation between the male coupling member **310** and the female coupling member **320** can be made only through a path through which the first inclined surface **311** of the male coupling member **310** moves along the second inclined surface **321** of the female coupling member **320**.

(34) The expansion end portion **314** may have a shape that allows its cross-sectional area to be modified by elasticity. Accordingly, when the expansion end portion **314** passes through the opening **323** of the female coupling member **320**, the cross-sectional area thereof is reduced and deformed by elasticity to enter the receiving space **S**, and the state in which the cross-sectional area thereof is expanded in the receiving space **S** may be recovered.

(35) Meanwhile, in an embodiment of the present invention, a portion of the extension portion **313** having the first inclined surface **311** may protrude outward than the expansion end portion **314** in a direction perpendicular to the extension direction of the extension portion **313**. In this way, when the first inclined surface **311** protrudes outward than the expansion end portion **314**, when an external force is applied in the first direction, the first inclined surface **311** may smoothly move along the second inclined surface **321** of the female coupling member **320** without interference from the expansion end portion **314**.

(36) Referring to FIGS. **4** and **6**, the female coupling member **320** may include a body **322** having a receiving space **S** to receive at least a portion of the extension portion **313** and the expansion end portion **314** and having a second inclined surface **321** on one side of the inner wall of the receiving space **S**, and an opening **323** formed in the body **322** in communication with the receiving space **S**. In this case, the second inclined surface **321** has a shape corresponding to the first inclined surface **311**.

(37) The opening **323** may include a first portion **323a** provided in a shape surrounding the extension **313** and a second portion **323b** connected to the first part **323a** and provided in a shape of a slot exposing the second inclined surface **321**. More specifically, in an embodiment of the present invention, the first portion **323a** may be formed in an arch shape, and the second portion **323b** may be formed in a rectangular shape.

(38) FIG. **7** is a diagram showing a process in which the first inclined surface of the male coupling member of the coupling member of the helmet moves along the second inclined surface of the

female coupling member to separate the male coupling member from the female coupling member according to an embodiment of the present invention. In addition, FIG. 8 is a diagram showing a state in which the male coupling member of the coupling member of the helmet is separated from the female coupling member according to an embodiment of the present invention. In FIGS. 7 and 8, the illustrations of the helmet body 100, the pad 200, and the cover plate 400 are omitted.

(39) Referring to FIGS. 7 and 8, in a state in which the male coupling member 310 and the female coupling member 320 are fastened in the coupling member 300 of the helmet according to an embodiment of the present invention, the expansion end portion 314 of the male coupling member 310 is fixed in the receiving space S of the female coupling member 320, and the expansion end portion 314 does not pass through the opening 323 of the female coupling member 320 unless the external force is applied in the first direction, that is, the direction in which the first inclined surface 311 can move along the second inclined surface 321. Accordingly, the male coupling member 310 and the female coupling member 320 are maintained in a coupled state, and the pad 200 is stably fixed to the inner surface of the helmet body 100.

(40) When an external force is applied in a state in which the male coupling member 310 and the female coupling member 320 are fastened in a first direction, i.e., a direction in which the first inclined surface 311 may move along the second inclined surface 321, the first inclined surface 311 may move along the second inclined surface 321 and the expansion end portion 314 may exit the receiving space S through the second portion 323b of the opening 323 of the female coupling member 320. As a result, the coupling state between the male coupling member 310 and the female coupling member 320 is released, and the pad 200 is separated from the helmet main body 100.

(41) In this regard, as shown in FIG. 9, the second inclined surface 321 and the second portion 323b may be formed to be wider than the expansion end portion 314. Specifically, the width W2 of the second inclined surface 321 and the second portion 323b of the opening 323 is formed to be larger than the width W1 of the expansion end portion 314 of the male coupling member 310. Therefore, when a force is applied in the first direction in the fastened state of the male coupling member 310 and the female coupling member 320, the first inclined surface 311 may move along the second inclined surface 321 and the expansion end portion 314 that has been fixed in the receiving space S may easily escape through the second portion 323b of the opening 323.

(42) Meanwhile, when the width W2 of the second inclined surface 321 and the second portion 323b of the opening 323 is larger than the width W1 of the expansion end portion 314 of the male coupling member 310, the expansion end portion 314 of the male coupling member 310 is introduced into the receiving space S through the second portion 323b of the opening 323 in a state in which the male coupling member 310 and the female coupling member 320 are separated, so that the fastening between the male coupling member 310 and the female coupling member 320 can be easily achieved.

(43) FIG. 10 is a diagram showing the arrangement and orientation of the female coupling member of the coupling member of the helmet according to an embodiment of the present invention.

(44) In one embodiment of the present invention, pads 200 are disposed on both left and right sides of the inner surface of the helmet body 100, respectively. In addition, a plurality of coupling member 300 are provided on both left and right sides to couple the helmet body 100 and the pad 200. In order to remove the pad 200, it is necessary to release the coupling state between the male coupling member 310 and the female coupling member 320 of each coupling member 300, which means that the first inclined surface 311 of the male coupling member 310 of each coupling member 300 must move along the second inclined surface 321 of the female coupling member 320. Therefore, it is preferable that the first inclined surface 311 and the second inclined surface 321 of each coupling member 300 are oriented so that each of the first inclined surface 311 can move along the second inclined surface 321 by a force in a single direction.

(45) Referring to FIG. 10, it can be confirmed that the second inclined surface 321 of each female coupling member 320 is oriented to face an extension line of an operation point of a force or an

operation direction of the force so that a coupling state of a plurality of coupling member **300** can be simultaneously released by the force (displayed by an arrow) acting downward in the drawing. Of course, the first inclined surface **311** of each male coupling member **310** is also disposed to correspond to the orientation of each second inclined surface **321**.

(46) FIG. **11** is a diagram showing a method of removing a pad in a helmet during an emergency according to an embodiment of the present invention.

(47) Referring to FIG. **11**, the helmet according to an embodiment of the present invention may include a pulling portion **210** provided to the pad **200** to be pulled by an external force in a first direction. More specifically, the pulling portion **210** is provided at the lower portion of the pad **200** so that the pad **200** can be pulled from the wearer's head toward the body. Here, the force in the first direction applied to cause the first inclined surface **311** of the male coupling member **310** of each coupling member **300** to move along the second inclined surface **321** of the female coupling member **320** becomes a force acting from the wearer's head toward the body.

(48) In this way, in an embodiment of the present invention, the first inclined surface **311** of the male coupling member **310** and the second inclined surface **321** of the female coupling member **320** may be oriented toward the pulling portion **210** or a virtual point extending in the first direction. Through this, the fastening state of the male coupling member **310** and the female coupling member **320** of the plurality of coupling member **300** can be simultaneously released through one external force. As a result, in an emergency situation, it is possible to easily separate the helmet from the head of the helmet wearer, and to prevent secondary injuries that may occur during the process of severely separating the helmet.

(49) The helmet according to the embodiment of the present invention may be basically applied to a riding helmet for a rider such as a two-wheeled vehicle and a race vehicle. However, the embodiments of the present invention may be applied to all kinds of helmets including helmets worn during sports games, such as ski, bobsley, and skeleton, but are not necessarily limited thereto.

(50) Although embodiments of the present disclosure have been described, the spirit of the present disclosure is not limited by the embodiments presented in the specification. Those skilled in the art who understand the spirit of the present disclosure will be able to easily suggest other embodiments by adding, changing, deleting, or adding components within the scope of the same spirit, but this will also be included within the scope of the spirit of the present disclosure.

Claims

1. A helmet, comprising: a helmet body; a pad coupled to an inside of the helmet body; and a coupling member that separably couples the helmet body and the pad, wherein the coupling member comprises: a male coupling member disposed in one of either the inside of the helmet body or the pad, and a female coupling member disposed in the other of either the inside of the helmet body or the pad and the male coupling member is inserted and fixed into the female coupling member, wherein the male coupling member has a first inclined surface at a portion inserted and fixed into the female coupling member, and the female coupling member has a second inclined surface corresponding to the first inclined surface, and when an external force is applied to the pad in a first direction so that the male coupling member is separated from the female coupling member in a state in which the male coupling member and the female coupling member are coupled, the first inclined surface moves along the second inclined surface, and the pad is separated from the helmet body, wherein the male coupling member comprises: a base portion fixed to either the inside of the helmet body or the pad; an extension portion that protrudes and extends from one side of the base portion and has the first inclined surface on one side thereof; and an expansion end portion provided at an end of the extension portion in an extension direction, at least a portion thereof extending outward than the extension portion in a direction perpendicular to the extension

direction, wherein the female coupling member comprises: a body having a receiving space to receive at least a portion of the extension portion and the expansion end portion and having the second inclined surface on one side of an inner wall of the receiving space, and an opening formed in the body in communication with the receiving space, wherein the expansion end portion passes through the second inclined surface and is inserted into the receiving space and is surrounded by the inner wall of the receiving space.

2. The helmet of claim 1, wherein a portion of the extension portion having the first inclined surface protrudes outward than the expansion end portion in a direction perpendicular to the extension direction.

3. The helmet of claim 1, further comprising a cover plate disposed to cover one surface of either the inside of the helmet body or the pad, and fixing the base portion.

4. The helmet of claim 3, wherein the cover plate has a through hole formed in a size that does not pass through the base portion, and wherein the base portion is disposed inside the helmet body or between the pad and the cover plate, and the extension portion extends through the through hole.

5. The helmet of claim 4, wherein the extension portion comprises: a first extension portion extending through the through hole, and a second extension portion disposed between the first extension portion and the expansion end portion and having the first inclined surface on one side thereof.

6. The helmet of claim 5, further comprising a movement prevention member the movement prevention member comprises: a fixing portion fixed to the cover plate, and a movement prevention portion connected to the fixing portion to prevent the movement of the male coupling member by being fastened to the extension portion.

7. The helmet of claim 6, wherein the second extension portion comprises a fastening portion to which the movement prevention member is coupled, and the movement prevention portion is coupled to the fastening portion.

8. The helmet of claim 1, wherein the opening comprises: a first portion provided in a shape surrounding the extension portion and a second portion connected to the first portion and provided in a shape of a slot exposing the second inclined surface.

9. The helmet of claim 8, wherein the second inclined surface and the second portion are formed to be wider than the expansion end portion.

10. The helmet of claim 1, further comprising: a pulling portion provided to the pad to be pulled by the external force in the first direction, and wherein the first inclined surface and the second inclined surface are oriented toward the pulling portion or a virtual point extending in the first direction.
